2026 UCHICAGO TRADING COMPETITION

Case Packet & Information

THE UNIVERSITY OF CHICAGO

Career Advancement
Financial Markets

THANK YOU TO OUR SPONSORS!

AQUATIC

Belvedere Trading

FLOW TRADERS

CTC
CHICAGO TRADING COMPANY

DRW

CITADEL | CITADEL | Securities

millennium

NTG

D E Shaw & Co

TREXQUANT

All Options

HRT

imc

Optiver

SIG
SUSQUEHANNA

Jane Street®

OLD MISSION

WELCOME

On behalf of the University of Chicago and the UChicago Financial Markets Program (FM), we are pleased to welcome you to the 14th Annual UChicago Trading Competition! We are excited to have you as part of our competition this year. Thank you to our Financial Markets Program student case writers and platform developers, and our corporate sponsors for their leadership and support.

This competition could not be possible without the generous support of our corporate sponsors. This year's sponsors include: DRW, DE Shaw, Belvedere Trading, Chicago Trading Company, Citadel, Flow Traders, Hudson River Trading, IMC, Jane Street, Old Mission Capital, Optiver, SIG, TransMarket Group, Millenium Partners, Trexquant, All Options, and Aquatic.

We are delighted to have DRW as a Platinum Sponsor this year!

The UChicago Trading Competition will be held in-person on Saturday, April 11th at Convene Willis Tower in downtown Chicago. The focus of this event will be algorithmic trading, with two cases covering the following themes:

1. Market Making
2. Portfolio Optimization

We are excited to have so many talented students in this year's competition, and we look forward to seeing the teams in action!

For more information about UChicago Career Advancement, visit our website.

TABLE OF CONTENTS

SCHEDULE OF EVENTS 1
PARTICIPANTS 2
AWARDS 2
COMPETITION TECHNOLOGY 3
CASE 1: Market Making 4
CASE 2: Portfolio Optimization 12

SCHEDULE OF EVENTS

Friday, April 10, 2026 (all times in CDT)

Location	Time (CDT)	Topic
Kimpton Gray Hotel *(122 W. Monroe)*	5:00 - 8:00 pm	Poker Tournament Sponsored by DRW

Saturday, April 11, 2026 (all times in CDT)

Location	Time	Topic
Convene - Willis Tower *(233 S. Wacker Drive)*	8:00 – 8:45 am	Breakfast and Participant Arrival
	8:45 – 9:00 am	Welcome and Agenda Overview
	9:00 – 9:15 am	Tech Case Prep
	9:15 – 12:15 pm	Case 1 Live Trading
	12:15 – 2:00 pm	Lunch and Employer Career Fair
	2:00 – 2:45 pm	Overview of Cases 1 & 2 and Q&A
	2:45 – 4:15 pm	Networking Reception
	4:15 – 5:00 pm	Awards Presentation

All sessions are required for participants to be eligible for prize money

PARTICIPANTS

We are pleased to announce that over 150 students across the United States will participate in this year's competition. The following institutions will be represented:

- Brown University
- California Institute of Technology (Caltech)
- Carnegie Mellon University
- Columbia University
- Cornell University
- Drexel University
- Emory University
- Georgia Institute of Technology
- Harvard University
- Johns Hopkins University
- Massachusetts Institute of Technology
- New Jersey Institute of Technology
- Northwestern University
- Princeton University
- Purdue University
- Rice University
- Stanford University
- University of Chicago
- University of Florida
- University of Illinois Urbana-Champaign
- University of Maryland
- University of Minnesota
- University of Pennsylvania
- University of Southern California
- University of Texas at Austin
- University of Washington
- University of California, Berkeley
- University of California, Los Angeles
- Vanderbilt University
- Williams College
- Yale University

AWARDS

Awards will be announced during the awards ceremony. Cash prizes will be awarded to the winning team of each individual case and the top three overall winners based on aggregate scores across all cases. **Participants must attend all sessions on Friday and Saturday to be eligible for prizes.**

COMPETITION TECHNOLOGY

The University of Chicago Financial Markets Program (FM) is excited to show its in-house trading platform X-Change built by members of the FM program for the 14th Annual UChicago Trading Competition! Case 1 will be run live utilizing this platform. Case 2 will be run before the competition and results will be played back at the event.

| Algorithm Development

Competitors may develop their algorithms in any computing language; however, Python will be the only officially supported language. No other languages will receive explicit support from the case writing team. On the day of the competition, one user from each team will be responsible for manually starting the team's algo at the beginning of each case round.

Additional details on rules and requirements for each round can be found in the case descriptions.

| Case Submission Dates

CASE 1

Competitors will run their finalized algorithms locally on the day of the competition.

CASE 2

Final code submitted by 11:59pm CST on Thursday, April 9.

This case will be run in advance of the competition. Teams will not run their algorithms live on the day of the competition. Final scores will be announced on the day of the competition.

CASE 1: Market Making

Introduction

In this case, you have been tasked to trade two stocks, one ETF, two prediction markets, and option contracts on a third stock (put/call). Your goal is to make as much money as you can, trading against other competitors and the bots that already exist on the exchange.

This case involves the stocks **A** (a small-cap stock) and **C** (a large-cap stock). We also provide call and put options contracts on stock **B** as well as the **ETF**, which consists of 1 share of each of the 3 stocks. Beyond the stocks, participants can trade on one prediction market regarding hypothetical Federal Reserve rates and one "meta" market which will be provided the day of the competition.

The case is structured as a series of successive rounds. Each round consists of 10 days, each day is 90 seconds long, and each second holds 5 ticks. Your positions hold over from day-to-day, and reset at the end of each round.

A is a traditional small-cap stock which releases earnings. You will receive their quarterly earnings as a structured news message twice per day, and based on a constant P/E ratio and the released earnings, you should calculate the new price of the stock.

B is a liquid semiconductor company, but you will not have any information regarding the stock itself, so it's advised to focus on the options contracts. There is one underlying path for **B** over each round, and at each tick, you will be provided quotes for a single-expiry European option chain across 3 strikes around spot.

C is more involved, directly interacting with the Fed rates prediction market. **C** is a massive insurance company whose price will be dictated by the weighted sum of two components: their business operations and their bond portfolio. For operations information, you will receive their quarterly earnings news messages twice per day (at 22 and 88 seconds) as a baseline. But **C**'s P/E ratio will NOT be constant; it will be inversely proportional to the expected bond yields, since larger bond yields make **C** less attractive to investors. The rough structure of this relationship is given below where γ is a sensitivity constant and y_t and y_0 represent bond yields at time t and 0.

$$P_t^{\text{op}} = EPS_t \cdot PE_t, \quad PE_t = PE_0 \cdot e^{-\gamma(y_t - y_0)}$$

The bond portfolio will also inversely depend on the yields. For realism, we use a Taylor expansion to characterize this relationship:

$$\Delta B_t \approx B_0 \left(-D\Delta y_t + \frac{1}{2} C(\Delta y_t)^2 \right)$$

Here, B represents the value of the bond portfolio, D/C represent duration and convexity constants, and y_t represents yields at time t. Combining the two components gives the final price for C,

$$P_t = EPS_t \cdot PE_t + \lambda \frac{\Delta B_t}{N} + \text{noise}$$

where N is the number of outstanding shares and λ is a weighting constant.

The Federal Reserve Prediction Market aims to predict what decision our hypothetical Fed will make at the end of April in terms of interest rates. There are 3 possibilities, they will either hike rates by 25 bps (basis points), keep rates the same, or cut rates by 25 bps. You will be provided quoted probabilities for each outcome, and to help you trade, you will also receive structured and unstructured news releases. The structured news will be in the form of Forecasted vs. Actual CPI prints; Actual higher than Forecasted indicates inflation and points toward rate hikes, and vice versa points toward cuts. The unstructured news will be in the form of news headlines that may or may not relate to the Fed's decision.

You should also use the quoted probabilities to calculate the expected policy change and convert that into yields using the following formula:

$$E[\Delta r_t] = (+25) q_t^{\text{hike}} + 0 \cdot q_t^{\text{hold}} + (-25) q_t^{\text{cut}}$$

$$y_t = y_0 + \beta_y E[\Delta r_t]$$

Education

ETFs are investment funds that are traded on stock exchanges, similar to individual stocks. They hold a diversified portfolio of assets, such as stocks, bonds, or commodities, and are designed to track the performance of a specific index or sector. In the real world, ETFs offer investors a way to gain exposure to a broad range of assets without having to buy each one individually, and they are known for their low cost.

In the context of ETFs, creation and redemption processes involve entering and unwinding swap agreements to manage the ETF's exposure to the underlying index. Here's a succinct description:

An ETF can be thought of as an agreement between parties on the exchange; one can swap (for a small fee!) between 1 share of the ETF for 1 share of A, B and C each. Similarly, one can make a trade in the opposite direction (once again, paying a small fee).

Meanwhile, an option is a contract that gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a fixed price, the strike price, on a specific expiration date. A call option gives the right to buy, while a put option gives the right to sell. A European option can only be exercised at expiration, unlike an American option which can be exercised at any time.

More specifically, Put-Call Parity (PCP) is a foundational no-arbitrage relationship that must hold between the prices of European calls and puts sharing the same strike and expiry:

$$C - P = S_0 - Ke^{-rT}$$

where C and P are the call and put prices, S is the current spot price, K is the strike, r is the risk-free rate, and T is time to expiration. Intuitively, a long call and short put at the same strike synthetically replicates a long forward position, so if the two sides of this equation diverge, a riskless profit is available.

Potential Strategies

Provide Liquidity: Offer continuous buy (bid) and sell (ask) quotes on the exchange to trade with participants looking to take a position. Your bid quote should be less than your ask quote (why is this?). Hint: think about the difference between your bid and your ask (known as the spread) as payment for taking on risk.

Manage Risk: Monitor and manage your exposure to market movements and potential losses. Note: strategies that produce consistent profits will score higher than those with a similar expected return and higher variance.

Strategic Adaptation: Navigate a market that includes participants that are both more and less informed that you. Consider which participants you want to trade against. Consider how trades you see on the exchange (especially from what looks like smart money) should change your fair value for assets.

Market Impact: Understand how your trades may affect market prices, in terms of pushing prices up or down. Hint: think about how your market impact varies depending on the overall market liquidity.

Understand News: How will news affect the prices of the underlying assets? Understand both structured and unstructured news, and from this develop a quantitative model which encapsulates your want to buy/sell. Fundamentally, every significant movement in the stock market is driven by some amount of news, and some amount of noise.

Put-Call Parity: If quoted calls and puts are mispriced, violating the equation listed previously, you can trade the mispriced leg against a synthetic replication of it using the other leg and the underlying forward.

Box Spread: A box is constructed by combining a bull call spread and bear put spread at the same two strikes $K_1 < K_2$. Its payoff should always be regardless of where the underlying settles, so after taking the risk-free rate into account, if the box is quoted above or below the theoretical present value, there exists arbitrage.

Note that different strategies will be more effective at different points of the competition. Make sure to adapt your sub-strategies between rounds.

Round Specification & Scoring

Rounds: The competition will consist of 3 hours of rounds, each 15 minutes long, with **increasing difficulty and increasingly weighted scoring**. Competitors are randomly assigned to an exchange each round through a round robin process to trade against a variety of other market participants.

There will be a settlement price for all assets at the end of each round. Each team's P&L will be calculated using these final prices and added to their cumulative total. Positions do not carry over between rounds. The settlement price for the ETFs is computed as the NAV, and everything is marked out to the fair value.

Difficulty will progress over the rounds, typically through opposing market makers **increasing their spreads, decreasing volume**, and the underlying assets becoming more volatile. Market makers in later rounds will be much quicker to respond to news than those in early rounds. Moreover, strategies that generate positive P&L at the start of the competition are expected to continue to generate positive P&L across rounds, but decrease over time. As a result, we want to encourage competitors to adapt as the competition progresses and edges become smaller, as later rounds become weighted more heavily.

We have a **nonlinear grading schema that converts P&L into points.** Strategies that generate consistent positive P&L are expected to do much better than strategies that are high risk/high reward in nature. Similarly, outlier results of both large positive and negative P&L will likely not excessively impact a team's total score. As such, a handful of terrible rounds will not erase a team's chance of overall success, nor will a handful of great rounds assure it.

Note that the practice round will not count towards total points.

Rules

You may take long or short positions in all assets available in a round, subject to risk limits specified below. Exceeding these risk limits will result in the rejection of your entire order.

Risk Limits

As a market maker, your firm stipulates the following risk assets across the tradable assets:

- Max Order Size
- Max Open Order (Size of unfilled order)
- Outstanding Volume (Total volume of unfilled order)
- Max Absolute Position (Sum of long and short positions)

If you exceed any of these positions, all further orders for that contract that exacerbate the risk limits will be blocked by the exchange. Similarly, if any order is rejected by limit, participants are not informed of which limit they breached. **We will release the specific risk limits in a pinned Ed post in advance of the competition. Risk limits are subject to change on the day of the competition.**

Miscellaneous Tips

In order to be successful in this case (and as a market maker in general), you will need:

- A market making algorithm that uses the predicted "fair value" to make profitable trades:
  - There will be some "smart" and "dumb" money bots on the exchange. Which bots should you want to trade against? Which bots should inform your future expectations of price/settlement price?
  - Is it always reasonable to quote symmetrically (in size and/or price) about your predicted fair price?
  - What modifications (if any) should you make in the way you trade if the fair price predicted by your model deviates from the mid-price of the market? What if only one side of your quotes gets filled?
  - When would it be justified to "cross the spread" to take a position?
  - Do you want to hold your positions to settlement? Does this expose you to additional risk?
  - Do you want to pay to get out of risk?
- ETF market making is a delicate task.
  - How can you identify arbitrage opportunities in ETF pricing in terms of redemption and creation?
  - Make sure to factor in redemption/creation costs!
  - Should you always take both sides of the arbitrage opportunity?
  - Hint: When the ETF and equity prices don't align, it's more likely that the ETF is mispriced.
  - If you're willing to take on more variance, it may be the case that the arb exists because the ETF is mispriced and the equity is fair, so if you do trade the arb, the trade on equity may have been bad.
  - When and why would one decide to swap short?
- Trading earnings is a difficult task. If you receive information before the rest of the market does, where will you be willing to buy and sell? Does it always make sense to post offers at the 'new price' or somewhere in between?

Code Submission

Competitors will however run their bots from a box provided to you on the actual day of the competition. Please make sure you know how to use VSCode and are able to SSH into a virtual box— should questions arise, we will be there to answer them.

Questions

For questions regarding Case 1, please post in the UChicago Trading Competition Ed in the "case1" folder.

We will regularly check for new messages.

CASE 2: Portfolio Optimization

| Introduction

Even as a child, playing in a kindergarten playground, you probably remember looking up to the sky and thinking: how do allocators construct a portfolio to maximize risk-adjusted returns? Fast-forward several years, and you now have a chance to explore this question.

In this case, you are a portfolio analyst tasked with allocating a fund across 25 assets grouped into five sectors. To aid you, your company has provided you with five years of historical data for each asset. Your goal is to develop an algorithm to construct and rebalance a portfolio aimed at maximizing returns while simultaneously minimizing returns variance.

You will be given a CSV with intraday prices (30 ticks per day). The algorithm you submit will be evaluated on the twelve months immediately after the period in the CSV.

| Education

SHARPE RATIO

Making money is good, but making it consistently is even better. The "Sharpe Ratio" attempts to capture this distinction by measuring how much return a portfolio earns per unit risk, and is given by

$$\text{Sharpe} = \sqrt{252} \cdot \frac{\text{mean}(r_t)}{\text{std}(r_t)}$$

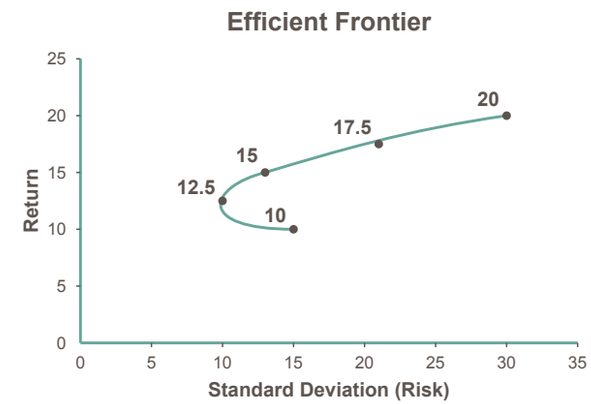
where r_t denotes the daily returns. We use the Sharpe ratio of your algorithm to compute your final score.

EQUAL WEIGHT PORTFOLIO

The naive portfolio allocation is giving each stock in your portfolio a weighting of $1/n$, assuming n distinct assets. This method assumes that your portfolio's assets aren't correlated with one another, and has many downsides. However, it's often a good baseline: if your approach is underperforming the equal weight portfolio, something's probably gone wrong. Our starter code implements this approach for you.

MARKOWITZ

In 1952, Harry Markowitz of the University of Chicago published "Portfolio Selection," which formalized this basic intuition and formed the basis of modern portfolio theory (MPT). Markowitz conceived of an efficient frontier of portfolios that maximize return given a certain level of risk (or alternatively minimize risk given a certain desired return). Along this efficient frontier, the actual portfolio chosen is based on the individual investor's level of risk aversion (high risk aversion = low level of risk, and vice versa). In the absence of exact knowledge on risk aversion, portfolio managers can choose an efficient frontier portfolio based on the specific objectives given to them by their clients or firms.



To implement a Markowitz portfolio, investors need some estimate of expected return and an estimate of risk. Additionally, we need some way to estimate the correlations, or covariances, between the returns of the different assets. Intuitively, a portfolio where all asset returns are highly correlated should have more risk than one with the same weights and risks for each individual asset but where all the asset returns are uncorrelated due to the fact that in the former case, one asset losing a large portion of its value means the entire portfolio likely will while this is not true in the latter case. Indeed, we see that the variance of returns in a portfolio (a measure of risk) is given by

$$\sigma_p^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \text{Cov}(r_i, r_j)$$

where w_i refers to the weight of the ith asset in the portfolio, r_i refers to the return of that asset, and n refers to the number of assets in the portfolio. While we can reasonably estimate variances and covariances of asset returns based on historical data, empirically, historical returns have not been a strong predictor of future expected return and large covariance matrix estimates tend to be numerically unstable, leading to practical difficulties in implementing a Markowitz portfolio. Since the development of MPT, investors and researchers have looked for new ways of dealing with this problem.

A downside of the Markowitz approach is that it works best on longer time horizons. Intraday price movements are often incredibly volatile and unpredictable— Markowitz based implementations are best used for reallocating at a 'macro' level rather than trading on a daily level.

RISK PARITY ALLOCATION (RPA)

One approach to portfolio allocation is to simply ignore expected returns when determining how to allocate assets in a portfolio. The most naive way of doing this would be to simply find the portfolio with the lowest predicted risk, but this would simply be to invest in a risk free asset, and the return offered by such a portfolio would not entice many investors. What we want instead is a way to have a portfolio full of risky assets where more weight is given to those with lower risk. This can be done by ensuring that every asset's risk contribution to the portfolio is equal, meaning that less risky assets have more weight than riskier ones in order to equalize the risk contribution. Here, risk contribution is calculated as:

$$\frac{w_i(\Sigma w)_i}{\sqrt{w^T \Sigma w}}$$

where w is the column vector of weights, and Σ is the variance-covariance matrix from the previous equation (not a summation of w). In this calculation, assets that are uncorrelated with the rest of the portfolio contribute less to risk than correlated assets with the same risk level, which is intuitive since a drop in the value of an uncorrelated asset does not contribute as much to overall losses for the portfolio as would a correlated asset. As it has been shown that asset volatilities and correlations are relatively stable over time, historical risk contribution can be used as a reliable estimate for risk contribution.

TOWARDS RETURN PREDICTION

Portfolio construction determines how to translate expected returns into weights. But where do the expected returns come from? In the simplest approaches, they are estimated from historical averages, which are insensitive to underlying structural patterns common in financial data. More sophisticated strategies attempt to forecast returns dynamically using these patterns.

Asset returns often exhibit structure that makes this possible: momentum effects, mean reversion, correlations across related assets, and clustering of volatility are all phenomena that researchers have documented and exploited. In this case, the sector structure of the universe is designed to reflect real economic relationships that influence how assets co-move and how they diverge. Signals that exploit patterns within and across sectors, or that identify periods when past returns are predictive of future ones, can compound into meaningful edges even when individual forecasts are weak. The most successful teams will go beyond static allocation and find ways to exploit predictable structure in the data.

Case Specification & Rules

The exchange trading platform will not be used for this case. Teams are expected to develop their strategies using our Python stub code and submit their code before the competition.

You will receive historical tick price data, a sector label, bid-ask spread (in basis points), and a borrow cost (annualized cost of holding a short position, in basis points) for all 25 assets. On each trading day, your algorithm will receive the price history observed so far and must submit an allocation. These allocations will be in the form of weights on each asset: weights can be positive, negative, or zero, but the sum of their absolute values must be at most one. If your algorithm outputs weights that violate this constraint, the evaluator will rescale them proportionally. Our starter code includes an example of how your code will be evaluated; this is being provided to help you understand the evaluation process only, and should NOT be taken as predictive of your final score – ignore this at your own peril!

Evaluation uses the full intraday price path rather than only daily closing prices. For each tick, compute the portfolio return by summing, across all 25 assets, each portfolio weight multiplied by that asset's simple return derived from its log return. Short positions also pay a borrow charge each tick equal to the sum, across assets, of the short exposure times the asset's annualized borrow cost times the fraction of a year represented by one tick. At the end of each day, when the portfolio is rebalanced from old weights to new weights, transaction costs depend on the change in each asset's weight and its bid-ask spread. The total cost is the sum of two parts: a linear term equal to half the spread times the absolute weight change, and a quadratic term equal to 2.5 times the spread times the squared weight change. Our starter code includes a reference evaluator so you can reproduce these mechanics locally.

You may use any packages, programming languages, and agents to study the training data we provide, but the submitted portfolio code must be in Python and will be restricted in dependencies. The default environment used to run submitted code will use Python 3.12 and will only have NumPy, pandas, scikit-learn, and SciPy installed. More details will be published on Ed and you will have the opportunity to request for additional packages.

We strongly advise that you test your submission using a similar environment on your local machine before submitting your final code; submitted code that does not compile or that fails to run for any time step will be disqualified for this case and the team that submitted it will receive 0 points. Before final submission, there will be an opportunity to test if your code compiles and runs properly (note that the Sharpes yielded from this test will not be indicative of what your actual Sharpe ratio will be).

Scoring

Teams will be ranked based on their annual Sharpe ratio over the 12-month test period.

Case Materials & Data

Python stub code and training data will be released along with the case packet and additional supplementary resources through the UChicago Trading Competition Ed.

We are requiring the final code for this case to be submitted by **11:59 PM CST on Thursday, April 9th**. Note that this is different from Case 1, as we will be computing the results of this round prior to the competition. Code submitted past this deadline will not be accepted, and we reserve the right to disqualify any competitors who submit incomplete code or miss this deadline.

Miscellaneous Tips

- Analyze returns, not prices. Prices of stocks tend to be non-stationary processes, but returns are generally stationary. Analyzing returns series will be more fruitful for your strategies than analyzing price series.
- Don't test strategies on the same data you train them on. Strategies will likely perform well on data your model has already seen – what's relevant is how well the strategy performs on data the model has not yet seen. You should not necessarily expect that your strategy will perform as well out-of-sample as it will in-sample; holding out a portion of your training data to test on (or running any other procedure to test on new data) is strongly advisable to get a more accurate sense of how successful your strategy will be. Every year, people overfit to the data – don't let that be you!
- Daily movement and intraday price changes are often two very different processes. Make sure you understand what the dynamics of each of these are—portfolio optimization is fundamentally about trading off short-term volatility with long-term predictability.
- Think carefully about costs. The scoring uses net returns after transaction costs and borrow fees. A strategy that rebalances frequently needs to outperform a lower-frequency one to justify the additional cost.

Questions

For questions regarding Case 2, please post in the UChicago Trading Competition Ed with the "Portfolio Optimization" tag.